

John Abrahams

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ML research engineer: reasoning grafts for frozen LLMs, new architectures rebuilt from the paper.

M.S.E. in Data Science, Penn, December 2026. Available remotely now; in person from December 15, 2026.

EDUCATION

University of Pennsylvania

M.S.E. in Data Science · GPA 3.62/4.0

Expected December 2026

Coursework: Theory of Machine Learning, Emerging Theory of Large Language Models, Advanced Operating Systems

Lehigh University

B.S. in Computer Science and Engineering · GPA 3.65/4.0 (major 3.75) · Minor: Probability and Statistics

May 2025

RESEARCH

Reasoning and Memory in Language Models, Independent research

Jul 2026 – Present

- Designed and built a plug-in recurrent reasoning module (HRM/TRM-style, with learned halting) that grafts into the reasoning band of a frozen LLM, training only about 5% of the parameters.
- Taught frozen Pythia-1B multi-step reasoning it could not do: eight-hop accuracy rose from near 0% to 54%, and reached 61% on entities never seen in training.
- Scaled the graft to Qwen2.5-Coder-7B, reaching 96% accuracy on eight-hop reasoning questions.
- Priced when a deployed LLM should learn a fact in its weights versus look it up: retrieval matched or beat weight updates on every test, and on Pythia-410M learning a fact pays off only after about 2,400 queries.

Recurrent-Depth and Von Neumann Architectures, Independent research

Sep 2026 – Present

- Reimplemented Geiping et al.'s recurrent-depth transformer (arXiv:2502.05171) from the paper and matched the released 3.5B Hugging checkpoint's logits to within $8.8e-05$ at every recurrence depth from 1 to 32.
- Reproduced its test-time compute scaling at fixed weights: as recurrence rose, SciQ accuracy climbed from 48% to 95% and ARC-Challenge from 26% to 38%, saturating where the paper reports.
- Rebuilt Chandra's Von Neumann Network (arXiv:2605.05780) from its equations alone and beat the paper's own MNIST result, 98.6% versus 96.4%, at the same 52.4K parameters.
- Redesigned the VNN to lift MQAR associative recall from 14% to 96.3% test accuracy, within seed noise of a parameter-matched recurrent-depth transformer (99.2%), with only 3,168 non-embedding parameters.

How LLMs Choose What to Compute, Independent research

Sep 2026 – Present

- Found a format-independent operation code inside Qwen3-8B: injecting it makes the model run a different operation on 87% of trials, and a rank-4 subspace carries it in both Qwen3-8B and OLMo-2-7B.

EXPERIENCE

Ninth Wave, Software Engineer Intern, AI Team

May 2026 – Aug 2026

- Built an infrastructure generator that stands up a new bank client's AWS stack from one short YAML file, replacing about ten hand edits per bank that had misrouted production traffic for all three banks.
- Built Studio, an MCP gateway giving AI agents one access-controlled entry point to the company's internal apps, and rewrote it in TypeScript to design out two authentication bypasses found in security review.
- Shipped 11 changes to the Spring Boot portal bank clients log into, including white-labeling and an S3 file browser.

New York Brain & Spine Surgery, Software Engineer (part-time, team of three)

May 2025 – Mar 2026

- Built the LLaMA and FastAPI backend for a clinical therapy chatbot prototype, with clinical rules steering every conversation and a conversation-state API that let the team change and test prompts quickly.

New York Brain & Spine Surgery, Research Associate

May 2023 – Aug 2023

- Co-authored a 236-patient spine-surgery outcomes study, modeling predictors of fair or poor outcomes.

PROJECTS

Kronos: a Type II exovisor, Independent project

Oct 2024 – Present

- Built and launched Kronos, a cloud where people and AI agents deploy Python or JavaScript programs that anyone can run from a link with no account; ran it in production from July 2026 with its first outside users.
- Wrote its Rust runtime, isolating every program in its own time-limited WebAssembly sandbox; a 12-page paper on the architecture measures about 0.15 MiB of memory per added sandbox.

SKILLS

Languages: Python, Rust, TypeScript, Java, C++, Go

Machine Learning: PyTorch, Hugging Face, activation patching, LoRA, retrieval, reinforcement learning

Systems: WebAssembly, AWS (CDK, Lambda, Bedrock, S3), Spring Boot, FastAPI, MCP, SQLite, Kubernetes